ALFA: A FRIENDLIER FACE FOR XACML

A major criticism of XACML has always been its verbose XML syntax, which can be cumbersome for developers to read and write. The Abbreviated Language for Authorization (ALFA, https://alfa.guide) was introduced to solve this problem, offering a compact, code-like syntax that compiles directly into standard XACML. ALFA maintains XACML's structure—policies, policy sets, rules, and combining algorithms—but replaces XML tags with a more familiar, programming-style format.

For example, instead of writing dozens of nested XML elements, the XACML policy we examined earlier could look like this in ALFA:

```
namespace acme
policy ConfidentialDocAccess {
    target clause resource.confidential == true;
    apply permit if principal.department == resource.owner.departm
ent;
}
```

The contrast shows how ALFA maintains XACML's semantics while greatly reducing verbosity. ALFA doesn't alter XACML's core model or semantics, but it makes authoring policies quicker and more user-friendly.

Originally developed by Axiomatics as the "Axiomatics Language for Authorization," ALFA was designed to make XACML policies more accessible to developers. In 2014, Axiomatics donated the language to the OASIS XACML Technical Committee, where it was standardized as "Abbreviated Language for Authorization." Today, ALFA is supported by the Axiomatics Policy Server and tools like a VS Code extension, which provides syntax highlighting, autocomplete, and direct compilation into XACML 3.0.

ALFA continues to serve as a helpful bridge for organizations committed to XACML, offering cleaner and more maintainable policy authoring without sacrificing compatibility with existing engines.